

TRANSPORT PHENOMENA I

FALL 2025

EXAM 1

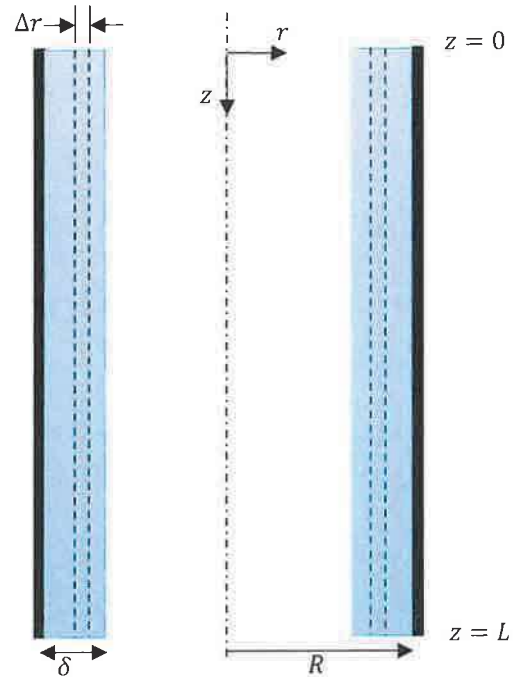


A thin liquid film of thickness δ is flowing down on the inner surface of a long vertical cylindrical tube of inner radius R .

Assumptions:

- Laminar flow
- Newtonian fluid with constant density ρ and constant viscosity μ
- Constant pressure within the liquid: $\frac{\partial p}{\partial z} = 0$
- Steady state flow, where $v_z = v_z(r)$ and $v_r = v_\theta = 0$

Answer the following questions, in terms of parameters δ , R , μ , ρ , and g . and L ?



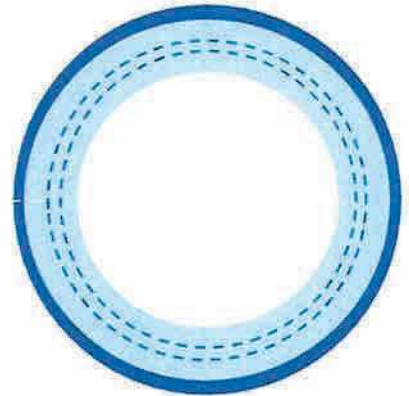
25/40 1- Write down the steady state z -momentum balance on a cylindrical shell of thickness Δr and length L (represented as the dashed lines in the schematics).

5/10 2- Apply Newton's law of viscosity to the above momentum balance to derive a second-order differential equation for the velocity profile of the liquid.

10/10 3- Write down two boundary conditions for this velocity profile.

10/20 4- Derive the velocity profile by integrating the differential equation of part 2 and evaluating the integration constants.

5/20 5- Calculate the mass flow rate of the liquid film.
Hint: $\int dx x \log x = \frac{1}{2} x^2 \log x - \frac{1}{4} x^2$



$$W = \rho V A \quad \text{where } V = \langle V_{avg} \rangle$$

$$\langle V_{avg} \rangle = \frac{\int_0^{2\pi} \int_\delta^R V_z r dr d\theta}{\int_0^{2\pi} \int_\delta^R r dr d\theta} = \frac{V_z \left(\frac{R^2}{2} - \frac{\delta^2}{2} \right)}{2\pi (R - \delta)} = \frac{V_z (R + \delta)(R - \delta)}{2(R - \delta)} = \frac{V_z (R + \delta)}{2}$$

Name: Aliya Kareemullah

1.) Momentum balance = 0 @ steady state

Convective momentum: $2\pi r \Delta r \left[\tau_{zr} \right]_{z=0} - 2\pi r \Delta r \left[\tau_{zr} \right]_{z=L}$

Molecular momentum: No pressure gradient. $\tau_{zr} = \tau_{rz} = -\mu \left[\frac{\partial v_z}{\partial r} + \frac{\partial v_r}{\partial z} \right]$ Newton's law of viscosity @ constant density $v_r = 0$

Surface area: $\delta L r$

$$\tau_{zr} L r \Big|_r - \tau_{zr} L r \Big|_{r+\Delta r}$$

Gravity forces: $-\rho g z$

$$0 = [2\pi r \Delta r]_{z=0} - [2\pi r \Delta r]_{z=L} + [\tau_{zr} L r]_r - [\tau_{zr} L r]_{r+\Delta r} - [\rho g z]$$

2.) Convective Flux = 0

$$[\tau_{zr} L r]_r - [\tau_{zr} L r]_{r+\Delta r} = \rho g L \quad z=L$$

Divide by volume

$$\frac{[\tau_{zr} r]_r - [\tau_{zr} r]_{r+\Delta r}}{r \Delta r} = \frac{\rho g L}{L r \Delta r} \quad \text{Take limit } \Delta r \rightarrow 0$$

$$\frac{1}{r} \frac{d\tau_{zr}}{dr} = \frac{\rho g}{r \Delta r}$$

$$\int \frac{d\tau_{zr}}{dr} = \int \frac{\rho g}{r \Delta r}$$

$$\tau_{zr} = \frac{\rho g}{\Delta r} r + C_1 \quad \tau_{zr} = -\mu \frac{\partial v_z}{\partial r}$$

$$\int \frac{\partial v_z}{\partial r} = \int -\frac{\rho g}{\Delta r \mu} r + \frac{C_1}{-\mu}$$

$$v_z = \frac{-\rho g}{2 \Delta r \mu} r^2 + \frac{C_1 r}{-\mu} + C_2$$

3.) $v_z(R) = 0$ ✓

$$\tau_{zr}(R-\delta) = 0 \quad \checkmark$$

$$4.) \tau_{zr} = 0 = \frac{\rho g}{\Delta r} (R-\delta) + C_1$$

$$C_1 = -\frac{\rho g}{\Delta r} (R-\delta)$$

$$v_z = 0 = \frac{-\rho g}{2 \Delta r \mu} R^2 + \frac{\rho g r (R-\delta)}{\Delta r \mu} + C_2$$

$$C_2 = \frac{\rho g R^2}{2 \Delta r \mu} - \frac{\rho g r (R-\delta)}{\Delta r \mu} = \frac{\rho g}{\Delta r \mu} \left(\frac{R^2}{2} - \frac{r(R-\delta)}{1} \right)$$

$$V_z = \frac{\rho g}{4\mu} \left(\frac{-r^2}{2} + (R-\delta) + \frac{R^2}{2} - (R-\delta) \right)$$

$$V_z(r) = \frac{\rho g}{4\mu} \left(\frac{R^2 - r^2}{2} \right)$$

$$5.) \quad w = \rho v A \quad A = 2\pi R L$$

$$v = \langle v_{avg} \rangle = \frac{V_z(R+\delta)}{2} \text{ (from front page)}$$

ρ is constant

$$w = \rho \times \pi R L \frac{V_z(R+\delta)}{2} = V_z \rho \pi R L (R+\delta)$$

$$w = \frac{\rho g}{4\mu} \left(\frac{R^2 - r^2}{2} \right) \rho \pi R L (R+\delta)$$

Name:

